

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Notional Process Design Data
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Notional Process Design Data

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Revision History

Rev No.	Date	Description of Revision
R01	02-07-2022	Issued for Review
R02	17-12-2022	Re-Issued for Review after incorporating TCE comments

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

The document, in consideration of relational design elements & concepts, forms the technical basis for the development of the facilities Project Specifications & tendering of contracts for the project's implementation.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission

ISPM	Industry Standard Process Modules
JV	Joint-Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

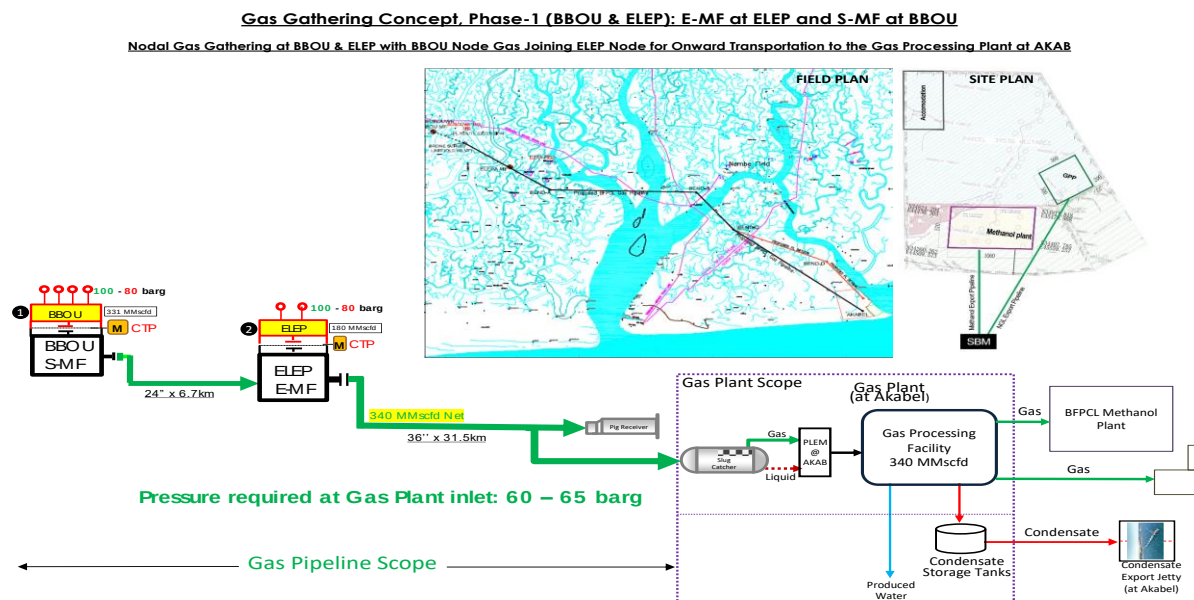


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose of Document

The purpose of this document is to provide the conceptual process design data for the development of the Project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

2.1. Concept Engineering Referenced Documents

Document Number	Title
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001- 3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.

2.2. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers. The design shall be based on DEP Feb 2019 (or latest, as advised by the EMPLOYER) some of which are listed below.

Document Number	Document Title
DEP 31.40.00.10-Gen.	Pipeline Engineering
DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems
DEP 31.40.10.14-Gen	Pipeline Overpressure Protection
DEP 31.40.21.33-Gen.	Pig signallers: Intrusive type
DEP 31.40.30.30-Gen	Concrete coating of line pipe
DEP 31.40.30.31-Gen	External polyethylene and polypropylene coating for line pipe

Document Number	Document Title
DEP 31.40.40.38-Gen	Hydrostatic pressure testing of new pipelines
DEP 31.40.50.30-Gen	Pre-commissioning of pipelines
DEP 31.40.60.11-Gen.	Pipeline leak detection

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. General Project Design Data

3.1. Design Units of Measurement

Variables	Units
Barg	Bar Gauge (Pressure)
Bara	Bar Absolute (Pressure)
bbl	Barrel (Oil Volume)
BCF	Billion Cubic Feet
Bopd	Barrels of Oil per day
Bpd or B/D	Barrel per day
BS&W	Base Sediment and Water
cP	Centipoise (Unit of Viscosity)
F	Fahrenheit
C	Centigrade
km	kilometre
m	meter
Mbopd	Thousand Barrels of Oil per Day
Mbpd	Thousand Barrels of per Day
Mbwpd	Thousand Barrels of Water per Day
MJ	Megajoule
mm	Millimetre
MMscfd	Million Standard Cubic Feet Per Day
MW	Mega-Watt
ppm	Part Per Million
psig	Pounds Per Square Inch Gauge (Pressure)
Component	Vol%, mol%, wt ppm
Liquid flow	Kg/h, m ³ /h
Steam flow	Kg/h
Level	%, m, mm
Density	Kg/m ³
Conductivity	μs/cm
Weight	Kg, t
Distance (length)	m, mm
Power	W, KW
Force	N, kN
Current	A, mA
Energy	MJ, kJ
Sound	dBA

Variables	Units
Frequency	HZ

3.2. Standard Process Conditions

Standard process conditions shall be taken as 15°C and 1.01325 bar.

3.3. Site and Environmental Conditions

The following site and environmental conditions have been extracted from Referencing paragraph 4.2.3 of Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited.

3.3.1. Air Temperature

Average maximum (daytime) temperature in the dry season months was about 32°C, and about 28°C in the rainy season. The minimum (night-time) temperature level was fairly the same (23°C) through-out the year.

3.3.2. Weather

Two weather conditions are prevalent in the region:

Dry Season (Harmattan)

During Harmattan season (November – January, 2-3 months/year), the atmosphere is laden with extremely fine dust (particle size of 5 microns or less), which can penetrate equipment, block air filters, etc.

Wet Season

Heavy rainfall can be expected in the wet season which normally runs from the months of April through October. Lightning is very frequent especially during the rainy season and shall be specifically considered in the design of all relevant equipment.

3.3.3. Rainfall

Rainfall steadily increases from about 65mm in January to about 288mm in July after which it drops to about 245mm in August. This amount increases to a second maximum peak of about 331mm in September and then steadily drops to 47mm in December. The double maxima characteristics of Equatorial climate are very evident in the project area.

3.3.4. Relative Humidity

Parameters	Values
Mean	81%
Minimum	69%
Maximum	88%

3.3.5. Wind Speed and Direction

The monthly mean wind speed in the project area varies from 6.2km/h to 9.6km/h with a mean of 8.5km/h.

The south-westerly wind is predominant (about 60% of the annual winds) and prevails during the wet season (March - October). The north-easterly wind which predominate during the dry season (November- February) and also brings the harmattan accounts to about 32% of the annual winds.

3.3.6. Seismic Zone

The project area is classified as seismic zone 0.

3.4. Design Life

All equipment for the project shall be designed based on a service life of 30 years.

4. Process Basis of Design

The pipeline shall be designed for multiphase flow due to the substantial amount of water and condensate in the constituent gas.

4.1. Project Location & Operational Conditions

The project location plan & operational conditions are summarized in the table below.

1.	Location	OML 33 & 35 - with the fields widely spread & on both sides of Nun River
2.	Target Fields Phase 1 starting fields (Subject of this BOD)	BBOU & ELEP in OML 33 - East of Nun River
3.	Target Fields Phase 1 subsequent fields	NUNR in OML 35, - East of Nun River SEIB in OML 35 - West of Nun River
4.	Target Fields Phase 2 fields	OPUG in OML 36 and OKPO in OML 35, - West of Nun River
5.	Desired Offtake Rate / GSPA Contractual Rate	340 MMscfd for 25 years / TBA
6.	Gas Offtake Points	SPDC Field Manifold Headers at the target fields, in phases starting with BBOU & ELEP, followed by NUNR, SEIB and OPUG
7.	Feed Stream Inlet Pressure to BFPC P/L System	80 barg – 100 barg
8.	Feed Stream Outlet Pressure from BFPC P/L System to Gas Plant	60 barg – 65 barg
9.	Gas Plant Processed Gas Outlet Pressure	41 barg – 49 barg
10.	Condensate rate	11 Mbcpd in year 1 to 34.4 Mbcpd in year 18.
11.	Water Production	Starting at 1.8 Mbwpd in year 1 and increasing to about 16.5 Mbwpd in year 14.

4.2. Production Forecast & Fluid Characteristics

Notional Fields Forecasts of Well Head Gas (WHG) Supply Required For Export Of 340 MMSCFPD Of Lean Gas To Methanol Plant For 20 Years With Elepa Development Accelerated to Year 1 OSD with Bubouwe Bou

4.2.1. BUBOUWE BOU

Condensate-Gas Ratio (CGR) is 5.56 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWP/D)
1	181	944	907
2	181	878	907
3	181	788	907
4	181	742	7855
5	181	707	7547
6	215	840	8491
7	305	1217	9002
8	327	1323	8193
9	331	1338	8038
10	67	239	2397
11	21	59	1031
12	16	42	784
13	2	4	85
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	800	3	21
	(BSCF)	(MMBBL)	(MMBBL)

4.2.2. ELEPA

Condensate-Gas Ratio (CGR) is 60.64 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWP/D)
1	180	10088	900
2	180	10088	900
3	180	10088	900
4	180	10088	7980
5	180	10088	7666
6	145	8146	4902
7	52	2911	2597
8	29	1630	1454
9	25	1386	1236
10	19	1076	960
11	11	628	561
12	9	478	426
13	1	52	46
14	0	0	0
15	0	0	0
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0
TCQ	435	24	11
	(BSCF)	(MMBBL)	(MMBBL)

4.2.3. NUN RIVER

Condensate-Gas Ratio (CGR) is 67.57 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWP/D)
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0
10	244	14464	1221
11	245	14583	1225
12	245	14583	1225
13	245	14583	10844
14	245	14583	10417
15	196	10914	6653
16	71	4159	3528
17	40	2329	1976
18	34	1980	1680
19	26	1538	1304
20	15	898	762
TCQ	586	35	15
	(BSCF)	(MMBBL)	(MMBBL)

4.2.4. SEIB+OPUG+ADD**

Condensate-Gas Ratio (CGR) is 65.63 BBL/MMscf.

Time (Year)	Gas (MMSCF/D)	Condensate (BCPD)	Water (BWPD)
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0
10	52	3049	259
11	108	6367	541
12	116	6831	580
13	139	8161	2857
14	141	8322	6096
15	188	11082	7496
16	308	18164	7717
17	341	24229	6297
18	348	32453	3334
19	354	31623	3116
20	363	31213	5551
TCQ	898	66	16
	(BSCF)	(MMBBL)	(MMBBL)

4.2.5. TOTAL SUPPLY PORTFOLIO

	Total WHG Supply*		CGR (BBL/MMscf) = 64.14
Time	GAS	CONDENSATE	WATER
year	MMSCF/D	BCPD	BWPD
Year	Gas	Condensate	Water
1	361	11032	1807
2	361	10965	1807
3	361	10876	1807
4	361	10830	15835
5	361	10794	15213

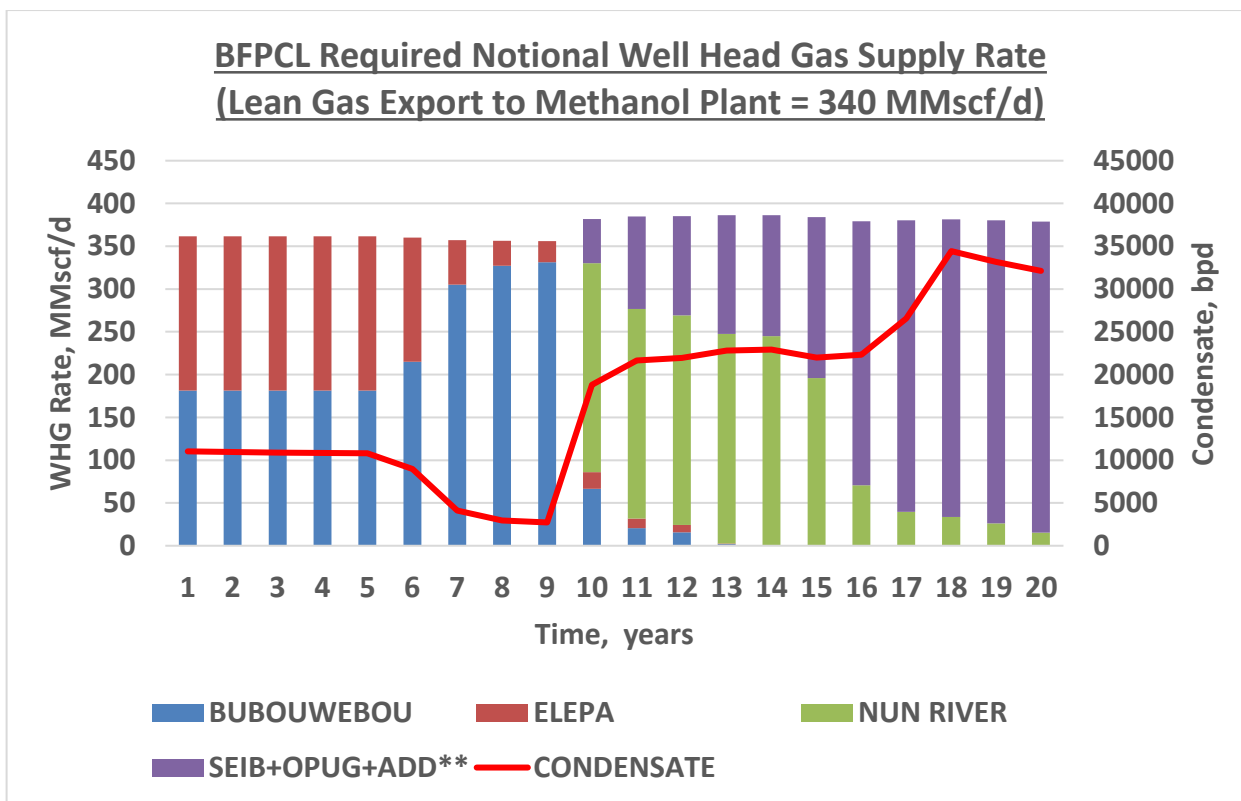
	Total WHG Supply*		CGR (BBL/MMscf) = 64.14
6	360	8985	13393
7	357	4128	11599
8	356	2953	9647
9	356	2724	9275
10	382	18828	4837
11	385	21637	3357
12	385	21933	3015
13	386	22799	13832
14	386	22905	16513
15	384	21996	14149
16	379	22323	11246
17	380	26558	8272
18	382	34433	5014
19	380	33161	4421
20	379	32111	6313
	2719	129	63
	(BSCF)	(MMBBL)	(MMBBL)

Notes

- WHG Supply* on Operating Day basis AND NOT Calendar day basis
- SEIB+OPUG+ADD** is subject to SPDC providing requested information
- Delivery Pressure range at Custody Transfer Point: 80 - 100 bar
- Delivery Temperature range at Custody Transfer Point: 50 - 95 degC

4.2.6. BFPCL Required Notional Well Head Gas Supply Rate

(Lean Gas Export to Methanol Plant = 340 MMscf/d)



4.2.7. Gas Compositions from Nominated Fields

4.2.7.1. Field Compositions

Field1: Bubouwe Bou Analogue		Field 2: Elepa		Field 3: Nun River H4.0	
Component	Mol.%	Component	Mol.%	Component	Mol.%
N ₂	0.07	N ₂	0	N ₂	NA
CO ₂	0.59	CO ₂	2.22	CO ₂	0.77
C ₁	92.92	C ₁	81.66	C ₁	72.83
C ₂	3.16	C ₂	6.13	C ₂	7.68
C ₃	1.01	C ₃	2.59	C ₃	4.85
iC ₄	0.61	iC ₄	0.75	iC ₄	1.46
nC ₄	0.44	nC ₄	0.81	nC ₄	1.97
iC ₅	0.25	iC ₅	0.45	iC ₅	0.95
nC ₅	0.19	nC ₅	0.33	nC ₅	0.76
C ₆	0.23	C ₆	0.65	C ₆	1.03
C ₇	0.19	C ₇	0.9	C ₇	1.41
C ₈	0.04	C ₈	0.87	C ₈	1.61
C ₉	0.03	C ₉	0.38	C ₉	0.9
C ₁₀	0.02	C ₁₀	0.3	C ₁₀	0.65

Field1: Bubouwe Bou Analogue		Field 2: Elepa		Field 3: Nun River H4.0	
C ₁₁	0.03	C ₁₁	0.23	C ₁₁	0.47
C ₁₂₊	0.22	C ₁₂	0.22	C ₁₂	0.37
C ₁₃	NA	C ₁₃	0.23	C ₁₃	0.34
C ₁₄	NA	C ₁₄	0.22	C ₁₄	0.3
C ₁₅	NA	C ₁₅	0.19	C ₁₅	0.26
C ₁₆	NA	C ₁₆	0.15	C ₁₆	0.21
C ₁₇	NA	C ₁₇	0.12	C ₁₇	0.16
C ₁₈	NA	C ₁₈	0.11	C ₁₈	0.14
C ₁₉	NA	C ₁₉	0.06	C ₁₉	0.08
C ₂₀₊	NA	C ₂₀₊	0.43	C ₂₀₊	0.8
TOTAL	100	TOTAL	100	TOTAL	100
Flash CGR	7	Flash CGR	59.7	Flash CGR	161.7

4.2.7.2. Portfolio Composition

Range for Portfolio	Lean Sample		Rich Sample
Component	Mol.%		Mol.%
N ₂	0.07		NA
CO ₂	0.59		0.77
C ₁	92.92		72.83
C ₂	3.16		7.68
C ₃	1.01		4.85
iC ₄	0.61		1.46
nC ₄	0.44		1.97
iC ₅	0.25		0.95
nC ₅	0.19		0.76
C ₆	0.23		1.03
C ₇	0.19		1.41
C ₈	0.04		1.61
C ₉	0.03		0.9
C ₁₀	0.02		0.65
C ₁₁	0.03		0.47
C ₁₂₊	0.22		0.37
C ₁₃	NA		0.34
C ₁₄	NA		0.3
C ₁₅	NA		0.26

Range for Portfolio	Lean Sample		Rich Sample
C ₁₆	NA		0.21
C ₁₇	NA		0.16
C ₁₈	NA		0.14
C ₁₉	NA		0.08
C ₂₀₊	NA		0.8
TOTAL	100		100

Notes

- STB - Stock Tank Barrel
- MMscf - Million standard cubic feet

4.2.7.3. Well Compositions

4.2.7.3.1. BBOU D3

As used in simulation model - Tuned PVTsim Model using Bonn-31ST PVT as analogue.

Component	Mol %	Molecular Weight	Liquid Density (g/cc)
N2	0.070	28.014	
CO2	0.590	44.010	
C1	92.922	16.043	
C2	3.160	30.070	
C3	1.010	44.097	
iC4	0.610	58.124	
nC4	0.440	58.124	
iC5	0.250	72.151	
nC5	0.190	72.151	
C6	0.230	86.178	0.664
C7	0.190	96.000	0.738
C8	0.040	107.000	0.765
C9	0.030	121.000	0.781
C10-C11	0.050	141.800	0.794
C12	0.068	161.000	0.800
C13	0.047	175.000	0.805
C14	0.032	190.000	0.808
C15	0.022	206.000	0.812
C16	0.015	222.000	0.815
C17	0.010	237.000	0.818
C18	0.007	251.000	0.821

Component	Mol %	Molecular Weight	Liquid Density (g/cc)
C19-C44	0.016	293.067	0.830
Total	100.000		

Bonny 31ST Reservoir Fluid Composition				
No	Component	Flash Gas	Flash Oil	Reservoir Fluid
		[mol %]	[mol %]	[mol %]
1	N ₂	0.07	0	0.07
2	CO ₂	0.59	0	0.59
3	C ₁	93.26	0	92.92
4	C ₂	3.17	0.01	3.16
5	C ₃	1.01	0.03	1.01
6	i-C ₄	0.61	0.07	0.61
7	n-C ₄	0.44	0.07	0.44
8	i-C ₅	0.25	0.16	0.25
9	n-C ₅	0.19	0.16	0.19
10	C ₆	0.22	1.88	0.23
11	C ₇	0.17	6.21	0.19
12	C ₈	0.02	6.61	0.04
13	C ₉	0	7.13	0.03
14	C ₁₀	0	5.7	0.02
15	C ₁₁	0	7.83	0.03
16	C ₁₂₊	0	64.14	0.22

Total		100	100	100
Density @ 60F	[g/cm ³]		0.811	
MW	[g/mol]	17.97	165.61	18.47
GOR	[Scf/Stb]			180,011.90
Gravity	[air=1]	0.62		
C ₇₊	Mole %	0.45	97.62	0.53
C ₇₊ MW	[g/mol]	100.89	167.69	143.17
C ₁₂₊ MW	[g/mol]	-	191.6	191.6
CGR (bbls/mmssc)				5.56

4.2.7.3.2. Elepa U-4

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
N2	0.0000		0.00		0.00
CO2	2.2200		0.00		2.30
H2S	0.0000		0.00		0.00
c1	81.6600		0.00		84.63
c2	6.1300		0.03		6.35
c3	2.5900		0.27		2.67
ic4	0.7500		0.37		0.76
nc4	0.8100		0.54		0.82
ic5	0.4500		0.87		0.43
nc5	0.3300		0.68		0.32
c6	0.6500		2.26		0.59
c7	0.9000		6.75		0.69
c8	0.8700		12.76		0.44
c9	0.3800		10.87		0.00
c10	0.3000		8.62		0.00
c11	0.2300		6.64		0.00
c12	0.2200		6.15		0.00
c13	0.2300		6.45		0.00
c14	0.2200		6.16		0.00
c15	0.1900		5.51		0.00
c16	0.1500		4.37		0.00
c17	0.1200		3.36		0.00
c18	0.1100		3.02		0.00
c19	0.0600		1.62		0.00
c20+	0.4300		12.70		0.00
Totals	100.00		100.00		100.00

4.2.7.3.3. Elepa U-7

Component	Res Fluid	Liquid		Gas	
	(z)		(x)		(y)
Totals	mol%		mol%		mol%
co2	2.1400		0.00		2.16
N	0.0000		0.00		0.00

Component	Res Fluid	Liquid		Gas	
c1	86.6700		0.00		87.30
c2	6.2100		0.37		6.25
c3	2.0500		0.24		2.06
ic4	0.5200		0.16		0.52
nc4	0.4900		0.26		0.49
ic5	0.2500		0.39		0.41
nc5	0.1600		0.36		
c6	0.2800		1.88		0.27
c7	0.3700		5.97		0.33
c8	0.3100		13.52		0.21
c9	0.1000		13.33		0.00
c10	0.0700		10.03		0.00
c11	0.0500		7.46		0.00
c12	0.0500		6.75		0.00
c13	0.0500		6.70		0.00
c14	0.0400		6.01		0.00
c15	0.0400		4.99		0.00
c16	0.0300		4.02		0.00
c17	0.0200		3.00		0.00
c18	0.0200		2.84		0.00
c19	0.0200		2.23		0.00
c20+	0.0700		9.49		0.00
Totals	100.01		100.00		100.00

4.3. Flow Assurance

A preliminary hydraulics study was carried out during the concept selection stage. A steady state hydraulic analysis shall be executed in the course of FEED phase to determine the following:

- Confirmation of the preliminary line sizes
- Optimise design / operating pressures for the pipelines.
- Establish expected flow regimes
- Liquid hold-ups, slug generated volumes and turndown capacities, and the impacts on slug management
- Evaluate sensitivities to line pack, flow rate ramp up / turn down and pigging operations
- Evaluate the impact of installation of new pipeline on the operability/maintainability of existing facilities.

The result is detailed in the Pipeline Hydraulic Study Report.

Steady State hydraulic analysis of the Multiphase Pipeline shall be carried out using PIPESIM Software. The results of the Pipeline hydraulic analysis shall be considered in the design of the pipeline system.

A transient hydraulic analysis shall be carried out during the FEED, and shall be incorporated into the start-up, shutdown and normal operations narratives of the Pipelines Operations and Maintenance Manual.

4.4. Materials for Construction

Line pipe materials for construction of the new pipeline sections shall be carbon steel API 5L Gr X70. with 3LPE and concrete coatings. The selection of materials shall consider the maximum and minimum process conditions, the high strain capacity requirements, all-seasons flooded swamp terrain, the extremes of ambient temperatures and all other conditions that are likely to affect the performance of the material. Particular attention shall be given to minimize the potential for brittle fracture of materials due to depressurization and shall be addressed in a pipeline depressurization / blowdown study carried out during detailed design. The pipeline shall be protected with chemical injection due to the CO₂ content of the gas.

4.5. Pipeline Design Parameters.

Design Condition	Value
Design Pressure	110 barg
Operating Pressure	100 barg
Min/Max. Design Temperature	12.7/80°C
Operating Temperature	50°C
Pipeline Depressurization Temp.	12.5°C
Blowdown line Depressurization Temp.	-13.3°C
Design Flow Rate	386 MMscfd

4.6. Equipment Description

4.6.1. Pig Launcher / Receiver

The first segment of the pipeline system; 24" x 6.7km Multiphase Pipeline from BBOU manifold to ELEPA manifold shall be fitted with a Pig Launcher and a Pig Receiver to facilitate its cleaning, inspection, and maintenance. The Pig Launcher shall be located at the BBOU manifold from where the pipeline starts while the Pig Receiver will be at the ELEPA manifold.

Also, the 36" x 31.5km segment of Multiphase Pipeline from ELEPA manifold to AKABEL shall be fitted with a Pig Launcher and a Pig Receiver to facilitate its cleaning, inspection, and

maintenance. The Pig Launcher shall be located at the ELEPA manifold while the Pig Receiver will be at the AKABEL Station.

The Pig Launcher / Pig Receiver shall be designed for use of intelligent pigs. Design code for the Pig Traps shall be ASME B31.8 while the End Closure shall be designed to ASME Section VIII Div. 1 and additional requirements of Shell DEP 31.40.10.13 – Gen. Technical Specification – Design of Pipeline Pig Trap Systems.

The quick opening closure shall be equipped with a mechanical interlocking system to prevent opening of the Pig Trap when pressurized. Pig signallers shall be installed on both sides of the pig launcher outlet valve to provide confirmation that the pig was successfully launched and passed through the pig trap valve. A Catch Tray shall be placed directly under the End Closure to prevent oil or debris contamination of the surrounding environment. Pressure Safety Valves shall be installed on the Pig Traps for overpressure protection.

Provision shall be made in the layout for Monorail Trolley for Pig handling operation at BOUBUWE BOU, ELEPA and AKABEL

4.6.2. Flare System

The vent from the Pig Launcher PSV, the Pig Launcher manual depressurisation line and the pipeline depressurisation line will be tied to a vent stack located around the manifold platform. A manual ball valve shall be used to enable the depressurization operation and the flow rate shall be controlled by a globe valve downstream of the ball valve. It is pertinent to state that the pipeline depressurization via a manual depressurisation line will be carried out at BBOU, ELEPA & AKAB end.

4.6.3. Drain Pan/Pit and Drainage System.

A Drain Pit, whose volume shall not be less than two times the volume of the Pig Trap shall be provided for the Pig receivers. It serves to collect drain from the Pig Traps drain pan install beneath the Pig trap door via gravity. The Drain Pit shall be covered to avoid rain water ingress and shall have a Level Gauge mounted on it. Provision for venting any trapped gas and for evacuation of the drain shall be made.

After pigging operation, inventory is evacuated from the pig trap through the vent and drain outlets. During normal operation, the pig trap is isolated from the process by closing all isolation valves upstream the trap. (Kicker line valve, balance line valve, pig trap isolation valve, vent/relief and drain valves). Positive isolation is achieved by means of a double block and bleed valve arrangement. The Drain Header is sloped towards the Drain Pit to enable complete draining of the Drain Header.

The drain pit at shall be evacuated by trucking/Vessel via a quick coupling connection mounted on the drain pit

4.7. Isolation Philosophy

The positive isolation of the Pig Traps and any pipeline segment shall be achieved by means of a double block and bleed valve arrangement together with a spectacle blind.

4.8. Pipeline Manual Depressurisation

To carry out maintenance activities on the Multiphase Pipeline system, the Pipeline system needs to be depressurized. To achieve this, a dedicated manual depressurisation line shall be provided at the manifold Stations and at Akabel. Suffice it to say that manual depressurization of the entire pipeline shall be carried out at Akabel station only. The manual depressurization line shall incorporate a Ball valve and a globe for flow control.